

Reticular Formation

Principles and Practices of Osteopathic Medicine I

- *Jennifer Xie, Ph.D.*
- Associate Professor
- Biomedical and Anatomical Sciences
- Arkansas, Wilson Hall, Rm # 144J

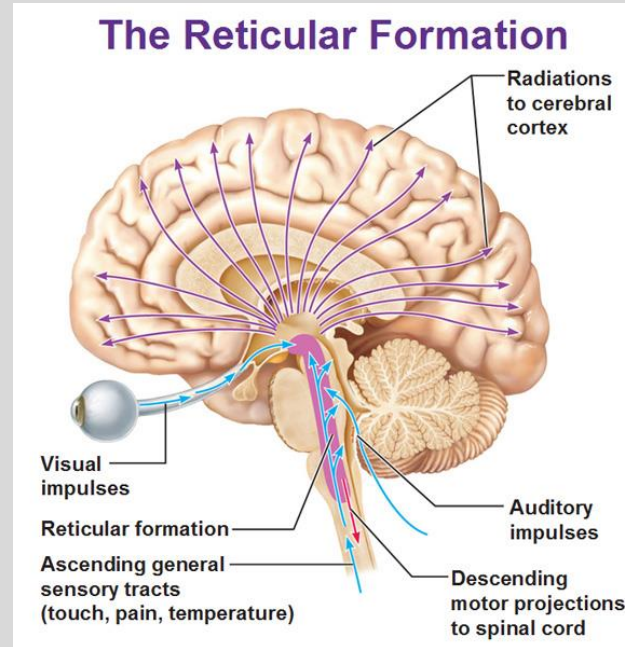
Email: Jennifer.xie@nyit.edu



Do.
Make.
Heal.
Innovate.
Reinvent the Future.

Session Objectives

- Describe the anatomical **organization** and divisions of the reticular formation
- Understand the major **functions** of the reticular formation
- Decipher the **neurochemistry** of the reticular formation
- Master the neurophysiology of **sleep**
- Summarize the clinical implications of reticular formation dysfunction



See **detailed objectives** in CPG

PLEASE READ THE LECTURE NOTES (SRX bricks not available)

Session Objectives Mapped to COMLEX Blueprint

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

- **Dimension 2**
 - **Clinical presentation 4.** Patient presentations related to the nervous system and mental health
 - 4.12 Sleep disturbances
 - Sleep disorders, including obstructive sleep apnea, somnambulism, insomnia, excessive daytime sleepiness, sleep-wake disorders, narcolepsy, night terrors, parasomnias
 - cognitive impairments, including altered level of consciousness, mild cognitive impairment, amnesia, coma, confusion, delirium, disorientation, subcortical and cortical dementia (eg, Alzheimer disease, Huntington disease, Parkinson disease)

Session Objectives Mapped to USMLE Blueprint

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

V. Nervous System & Special Senses

A. Normal Processes

2. Organ structure and function

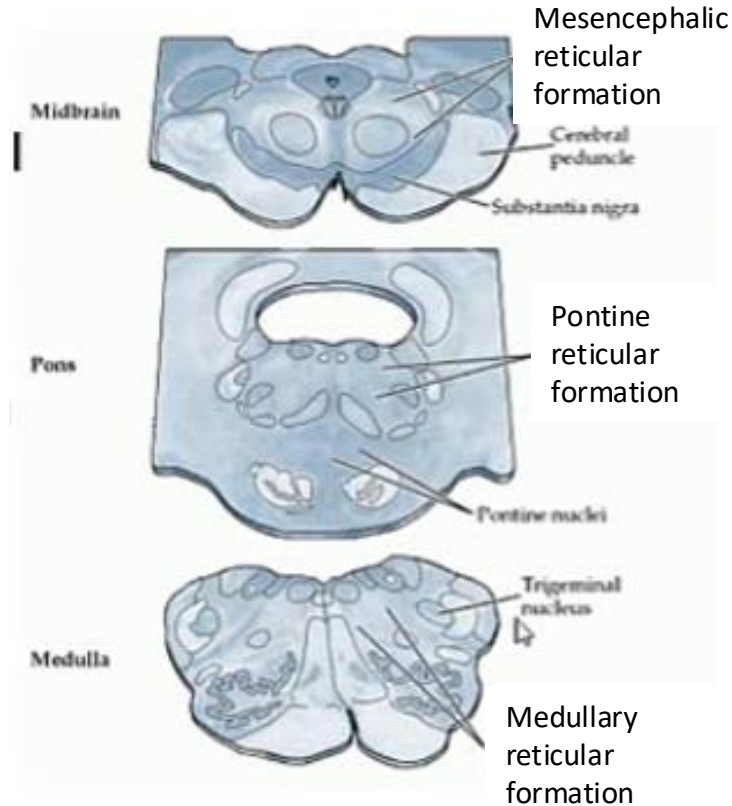
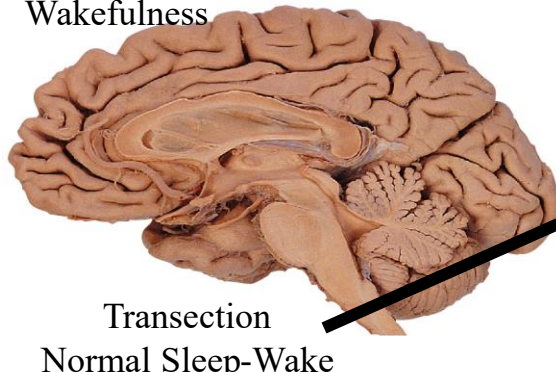
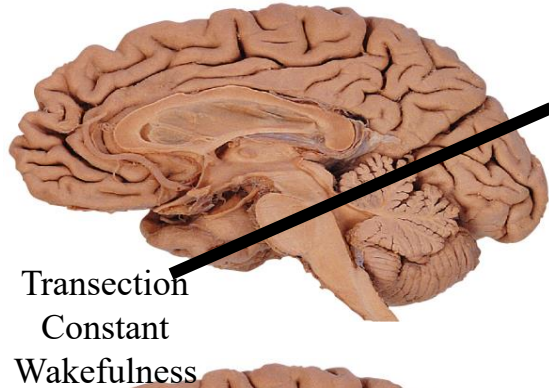
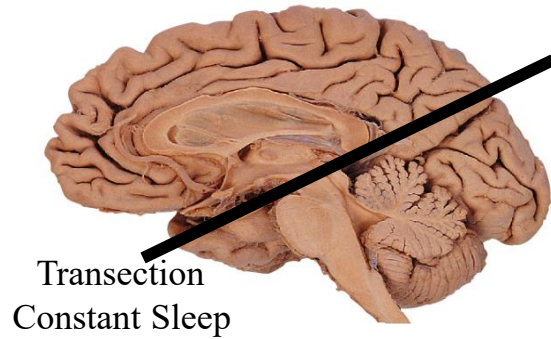
- ii. Brain stem (eg, cranial nerves and nuclei, reticular formation, anatomy and blood supply, control of eye movements)

B. Abnormal Processes: Health and Health Maintenance, Screening, Diagnosis, Management, Risks, Prognosis

5. Cranial and peripheral nerve disorders

- v. Global cerebral dysfunction: altered states of consciousness; delirium; coma/brain death
- x. Sleep disorders: cataplexy and narcolepsy; circadian rhythm sleep-wake disorder; insomnia, primary; sleep terror disorder and sleepwalking; REM sleep behavior disorder; restless legs syndrome

What is reticular formation?



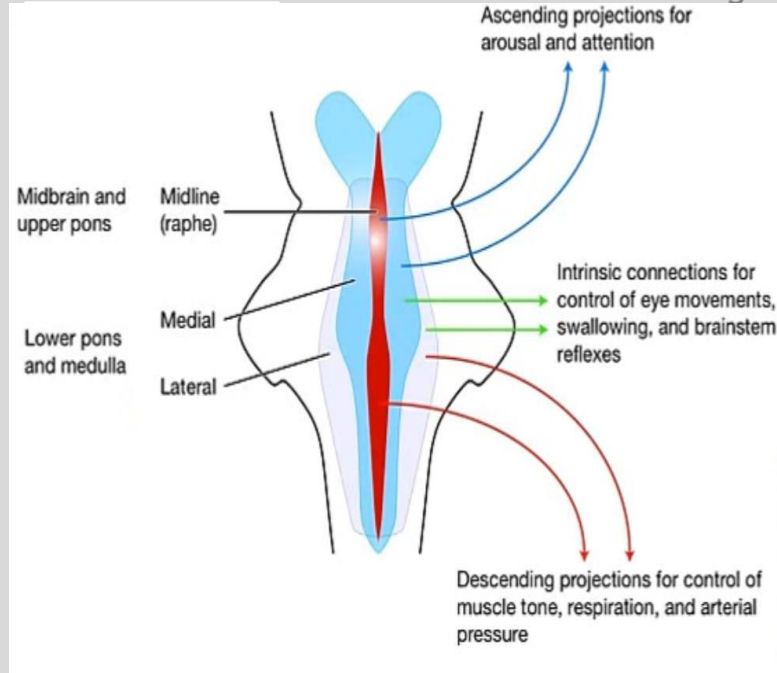
Reticular activating system discovered in cats in 1949, Moruzzi and Magoun, *Electroencephalography and Clinical Neurophysiology*

What is the Reticular Formation?

NEW YORK INSTITUTE
OF TECHNOLOGY

College of Osteopathic

- **No definitive definition**
- Consists of
 - Extended rostrally to the lateral hypothalamus and subthalamic nuclei; Caudally to the spinal cord grey matter
 - Short **interneurons** for **brain stem reflexes**
 - Long **projection neurons** with cell bodies in the brain stem & hypothalamus and their axons **ascending** to the **forebrain** or **descending** to the **spinal cord**
- Do **not** consist the cranial nerve nuclei or the long tracks passing through the brain stem



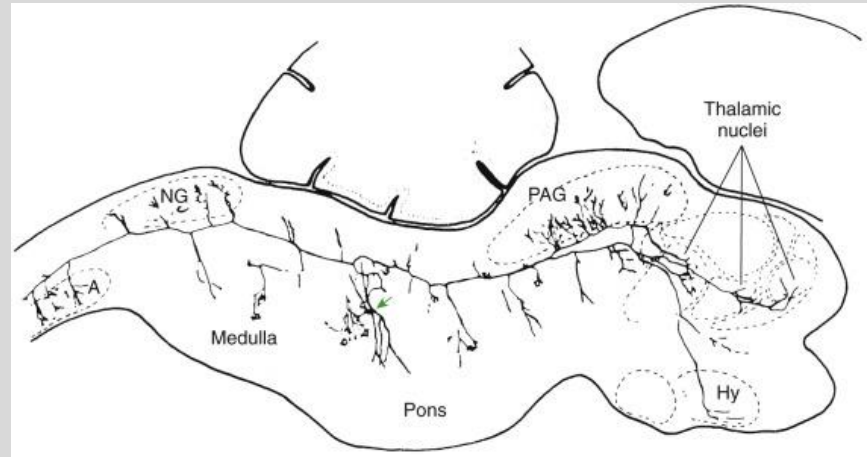
Nuclei of the Reticular Formation

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

Projection neurons release the following neurotransmitters to send **diffuse, modulatory** inputs to cerebral cortex, thalamus, hypothalamus and spinal cord:

- **Serotonin** (5-HT)
- **Norepinephrine** (NE)/epinephrine
- **Dopamine** (DA)
- **Acetylcholine** (ACh)
- **Histamine**



Diffuse projection system:

It is estimated that a single neuron in this network may have synapses with as many as 25,000 other neurons.

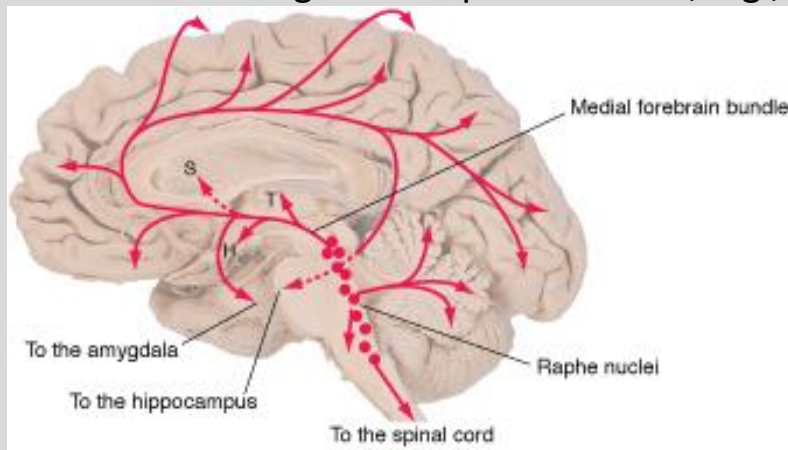
Serotonergic System

Cell bodies (nuclei):

- **Raphe nuclei:** along the entire brainstem
 - **dorsal raphe nuclei (midbrain)**, project to the **forebrain**
 - **raphe magnus nucleus (medulla)**, project to the **brainstem and spinal cord**

Functions:

- **Arousal & selective attention** (reticular activating system, **RAS**)
- **Sleep**
- Sensory processing, e.g., **pain**
- Regulation of muscle tone & segmental spinal reflexes, e.g., **reticulospinal system**



Noradrenergic System

NEW YORK INSTITUTE
OF TECHNOLOGY

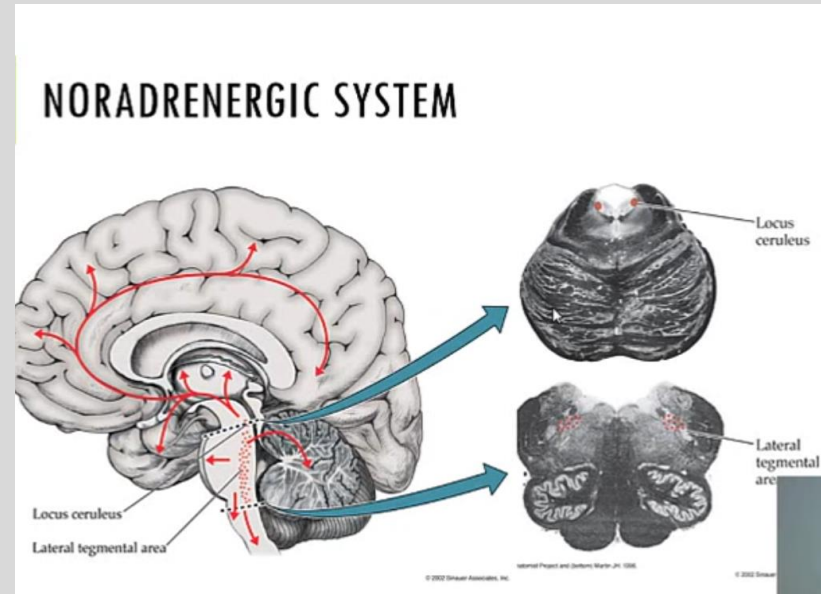
College of Osteopathic
Medicine

Cell bodies (nuclei):

- **Locus coeruleus**, project to all parts of the CNS
- Lateral tegmental area

Functions:

- **Arousal** & selective attention (reticular activating system, **RAS**)
- **Sleep**
- Sensory processing, e.g., **pain**
- **Autonomic** functions



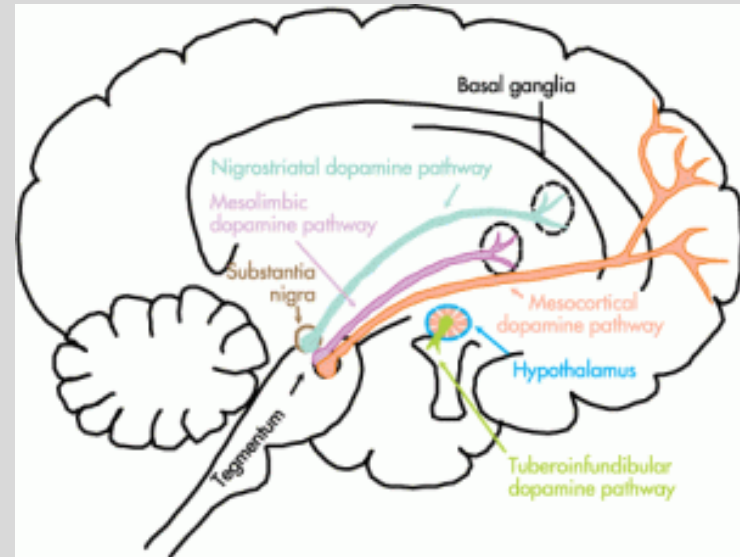
Dopaminergic System

NEW YORK INSTITUTE
OF TECHNOLOGY

College of Osteopathic
Medicine

Pathways - Functions:

- **Ventral tegmental area (VTA)**, mesencephalon
 - **Mesolimbic** – addiction, positive symptoms of schizophrenia, e.g., hallucinations
 - **Mesocortical** – cognitive deficits and hypokinesia in Parkinson's disease (PD), and “negative” symptoms of schizophrenia
- **Nigro-striatal system**
 - Originate in **substantia nigra pars compacta** – coordinating movements, major pathway related to PD
- **Tubero-infundibular system** - **Hypothalamus**, inhibit **prolactin** release from pituitary gland



Copyright © American Psychiatric Publishing, Inc., or American Psychiatric Association, unless otherwise indicated in figure legend. All rights reserved.

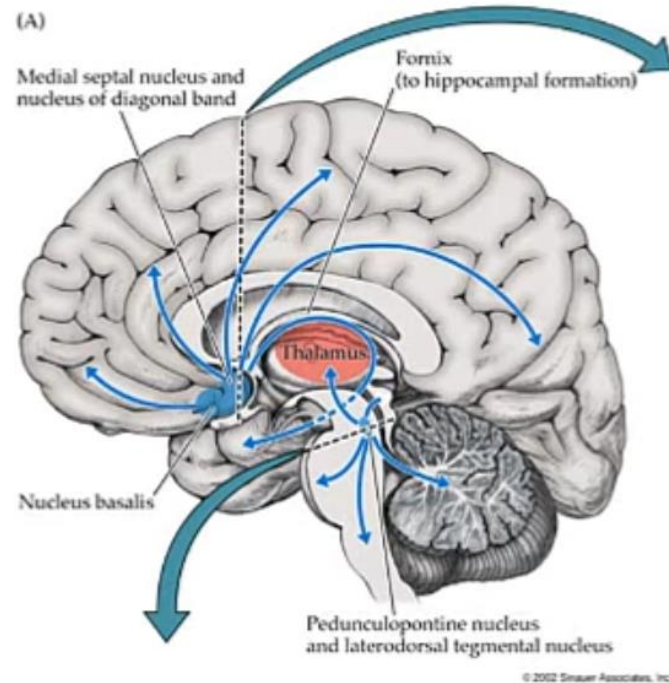
Cholinergic System

Cell bodies:

- **Nucleus basalis of Meynert (NBM, basal forebrain system)**, project to the entire cortex
- **Pedunculopontine tegmental nuclei (PPT)**, project to brainstem and spinal cord
- NBM & PPT activate cortical neurons via thalamic neurons

Functions:

- **Learning & memory, cognition - NBM (Alzheimer's Disease, Parkinson's Disease)**
- **Arousal (RAS) and selective attention**
- **Locomotion - PPT (fixed gait in PD)**



Histaminergic System

NEW YORK INSTITUTE
OF TECHNOLOGY

College of Osteopathic
Medicine

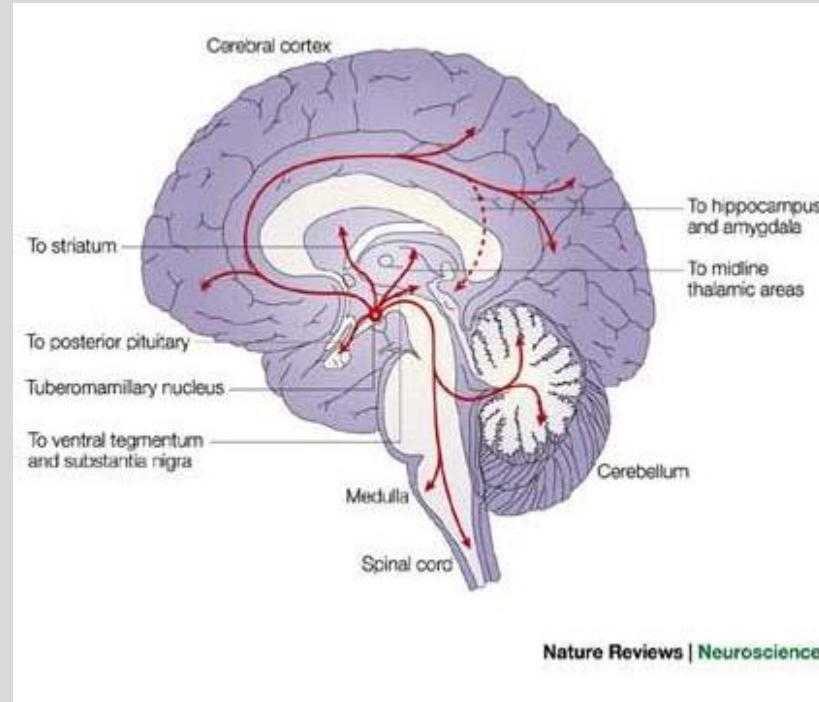
Cell bodies:

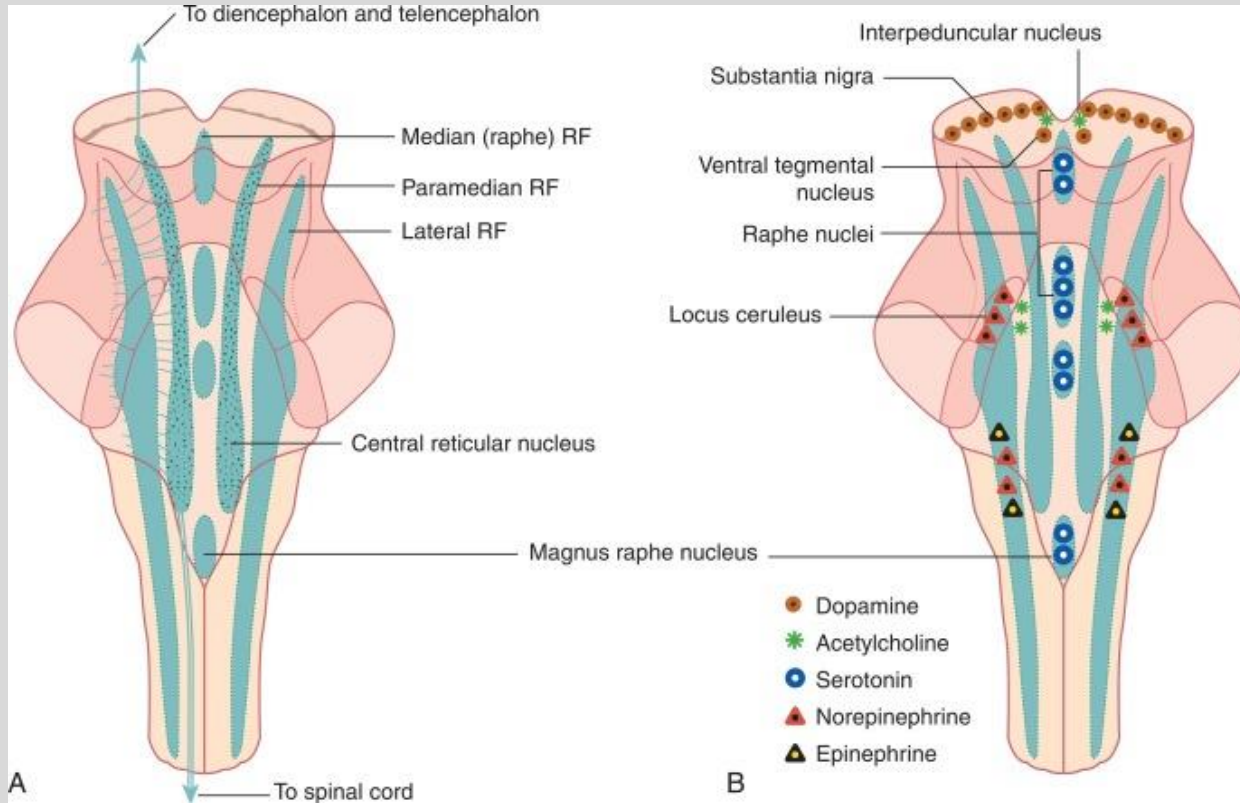
- **Posterior lateral hypothalamus** and **tuberomammillary** nucleus, project to all parts of CNS

Functions:

- **Arousal (RAS)**
- **Sleep**

(Think about what Benadryl can do)





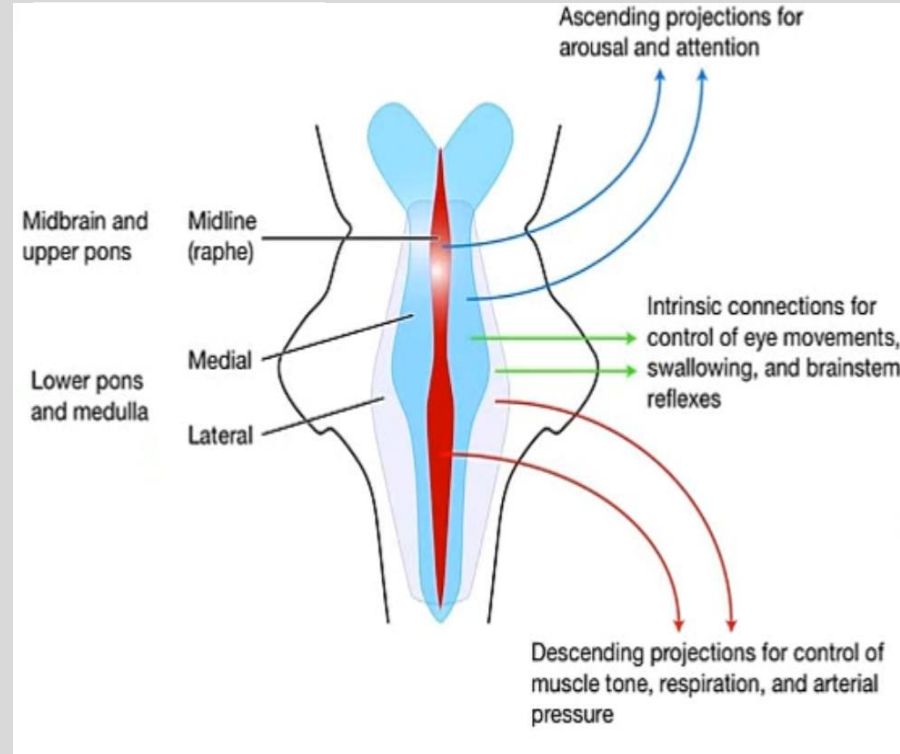
Functions of Reticular Formation

Ascending:

- Arousal
- Attention
- Sleep

Descending:

- Motor Functions
- Pain
- Micturition
- Cardiac Reflexes
- Respiration



Functions of Reticular Formation

**NEW YORK INSTITUTE
OF TECHNOLOGY**

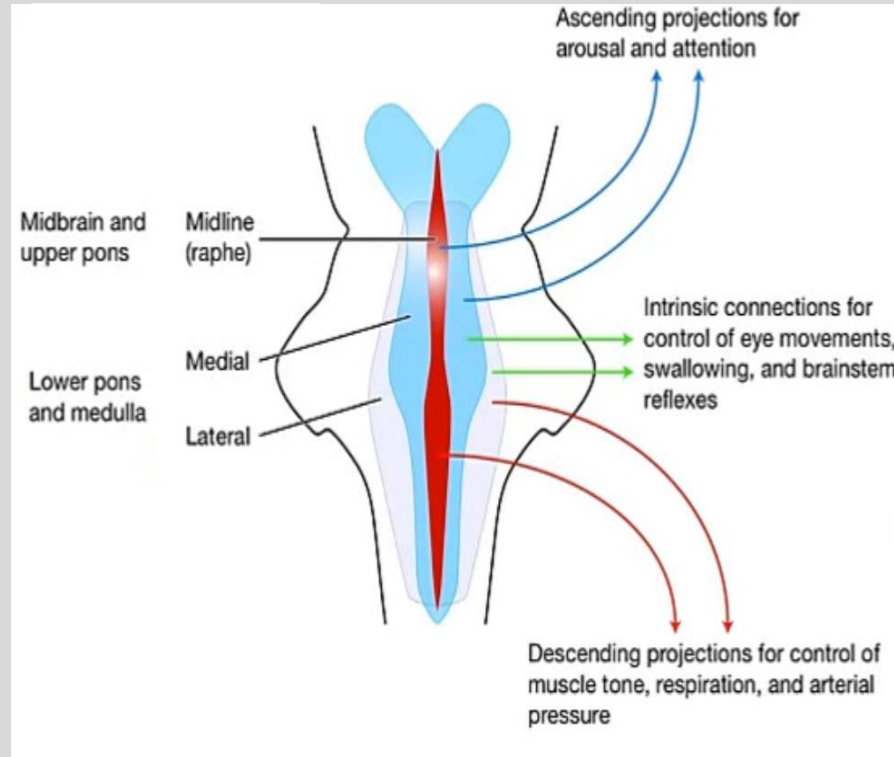
College of Osteopathic
Medicine

Ascending:

- **Arousal**
- **Attention**
- **Sleep**

Descending:

- **Motor Functions**
- **Micturition**
- **Pain**
- **Cardiac Reflexes**
- **Respiration**



Ascending Reticular Activating System (ARAS or RAS)

Consciousness:

- **Awake/alert**: depending on the ARAS
- **Aware**: depending on the cerebral cortex
- **All 5 projection neurons** are sending projection fibers up to the cortex to keep us awake and alert

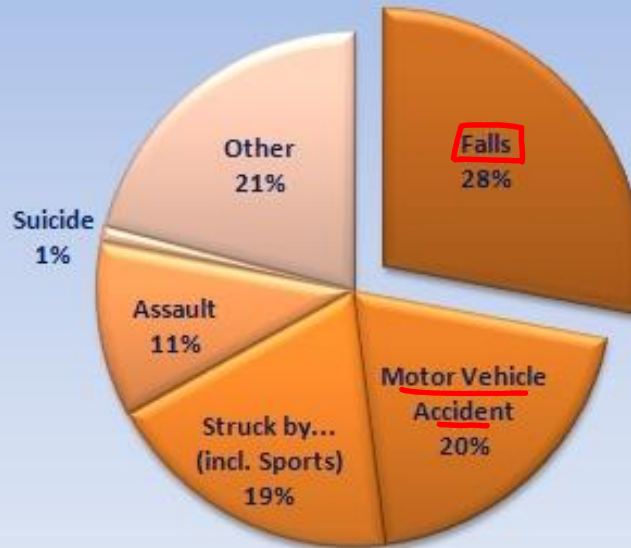
Traumatic Brain Injury

NEW YORK INSTITUTE
OF TECHNOLOGY

College of Osteopathic
Medicine

Traumatic Injuries

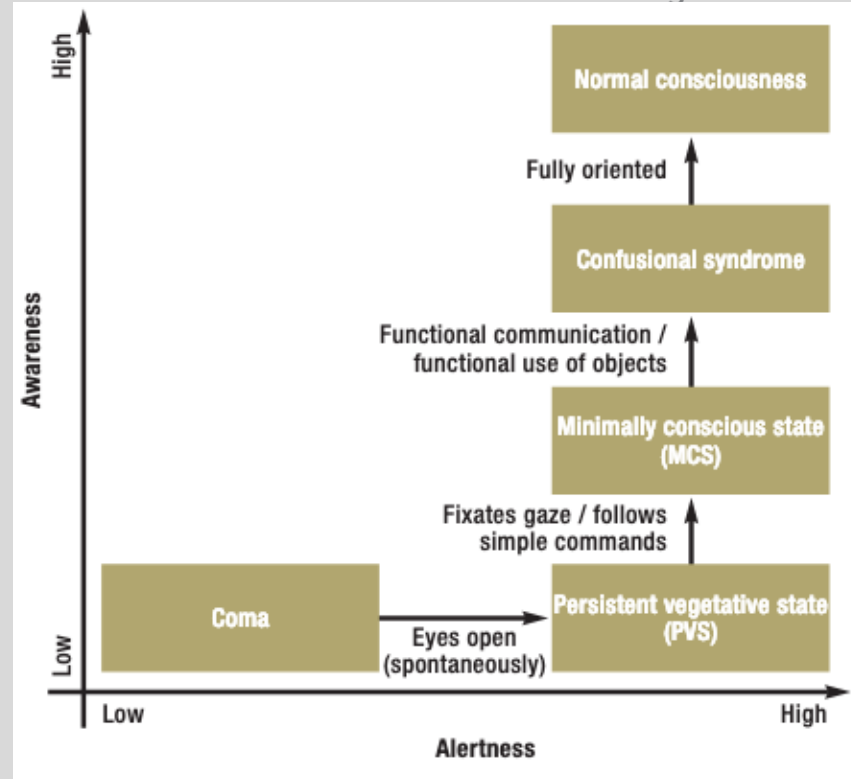
Major Causes of Traumatic Brain Injuries



Source: National Center for Injury Prevention and Control, CDC

Brain Injuries: Levels of Consciousness

- Coma
- Persistent Vegetative State
- Minimally Conscious
- Locked-in Syndrome



Coma: is marked by

- **Low arousal, low responsive** state
- Absence of eye, verbal or motor responses
- Abnormal EEG
- No sleep-wake cycle
- Needs respiration support
- Requires extensive lesions of
 - ARAS (most often) **or**
 - Both cerebral hemispheres (less common)
- **Glasgow Coma Scale (GCS)**: simple and reliable to gauge of the severity

GLASGOW COMA SCALE

Eye Opening

Spontaneous	4
To loud voice	3
To Pain	2
None	1

Verbal Response

Oriented	5
Confused /Disoriented	4
Inappropriate words	3
Incomprehensible sounds	2
None	1

Best Motor Response

Obeys	6
Localizes	5
Withdraws (flexion)	4
Abnormal flexion	3
Extension	2
None	1

*Normal Score = 15

*Mild Head injury = 14-15

*Moderate Head injury = 9-13

*Severe Head injury = less
than or equal to 8

Lowest score = 3 (not 0)

Persistent vegetative state

("unresponsive wakefulness") is marked by **moderate arousal** and **low awareness**.

- **Respond to external stimuli**, such as food and touch
- Eyes open and close
- Intact sleep-wake cycle
- Capable of swallowing
- Sometimes show very simple behaviors, teeth grinding, eye tracking, but do not show external evidence of consciousness
- Do **not** need artificial respiration
- Rarely return to normal consciousness if prolonged by more than a year

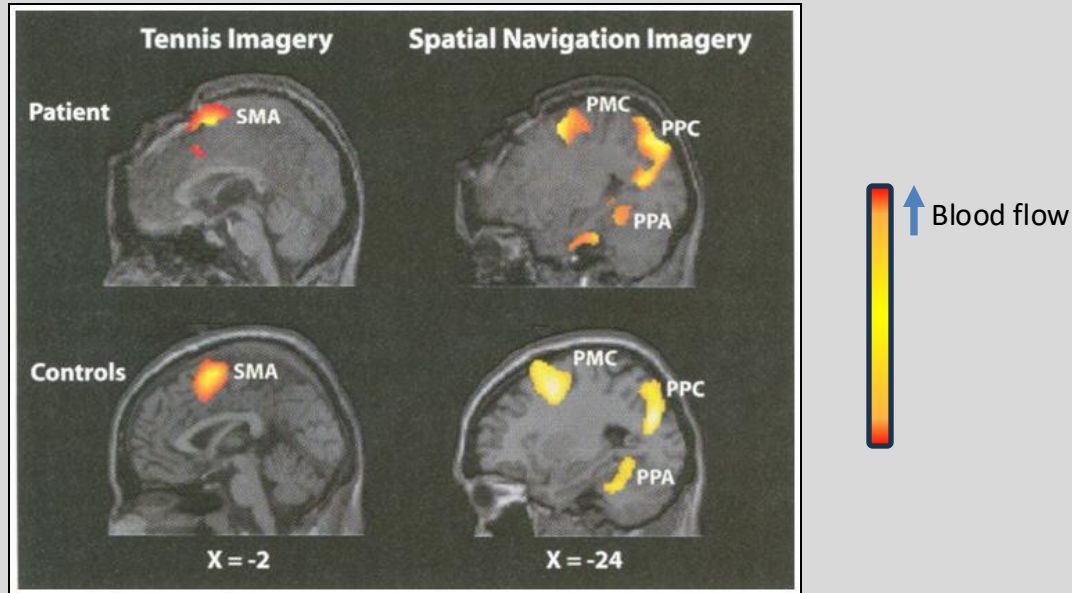


Terry Schiavo (1963-2005)

Additional brain processing during vegetative state revealed by fMRI

NEW YORK INSTITUTE
OF TECHNOLOGY

College of Osteopathic
Medicine



Detecting Awareness in the Vegetative State.

Owen, Adrian; Coleman, Martin; Boly, Melanie;
Davis, Matthew; Laureys, Steven; Pickard, John

Science. 313(5792):1402, September 8, 2006.

Fig. 1 BOLD imaging (to measure blood flow) showed activation of different brain areas when the patient was asked to imagine different types of activities. When the patient was asked to imagine playing tennis, supplementary motor area (SMA) lit up; in contrast, when the patient was asked to imagine going around the house, the parahippocampal gyrus (PPA), posterior parietal-lobe (PPC), and lateral premotor cortex (PMC) lit up. These patterns were similar to healthy volunteers, suggesting this vegetative state patient maintained awareness.

Minimally conscious state (new diagnosis) is marked by

- **Moderate arousal** and **intermittent awareness**.
- Similar to persistent vegetative state but there are **occasional external indications of consciousness**.
- Sometime response to instructions, some language, some appropriate recognition of objects.
- More consistent and steady improvement compared to persistent vegetative state.
- Rare but possible full recovery.

Locked-in syndrome is marked by

- **ARAS is spared** - full arousal and full awareness, but no voluntary movement aside from eye movements.
- **Bilateral artery stroke:** Dense pontine lesion severing motor tracts
- **Intact sensory** transmission
- Can use various devices to communicate (e.g. triggered by blinking, or brain-waves)
- Permanent

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic
Medicine

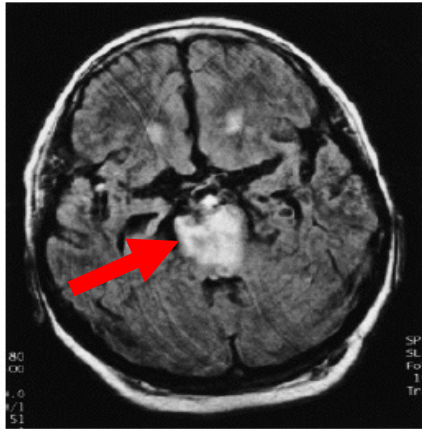


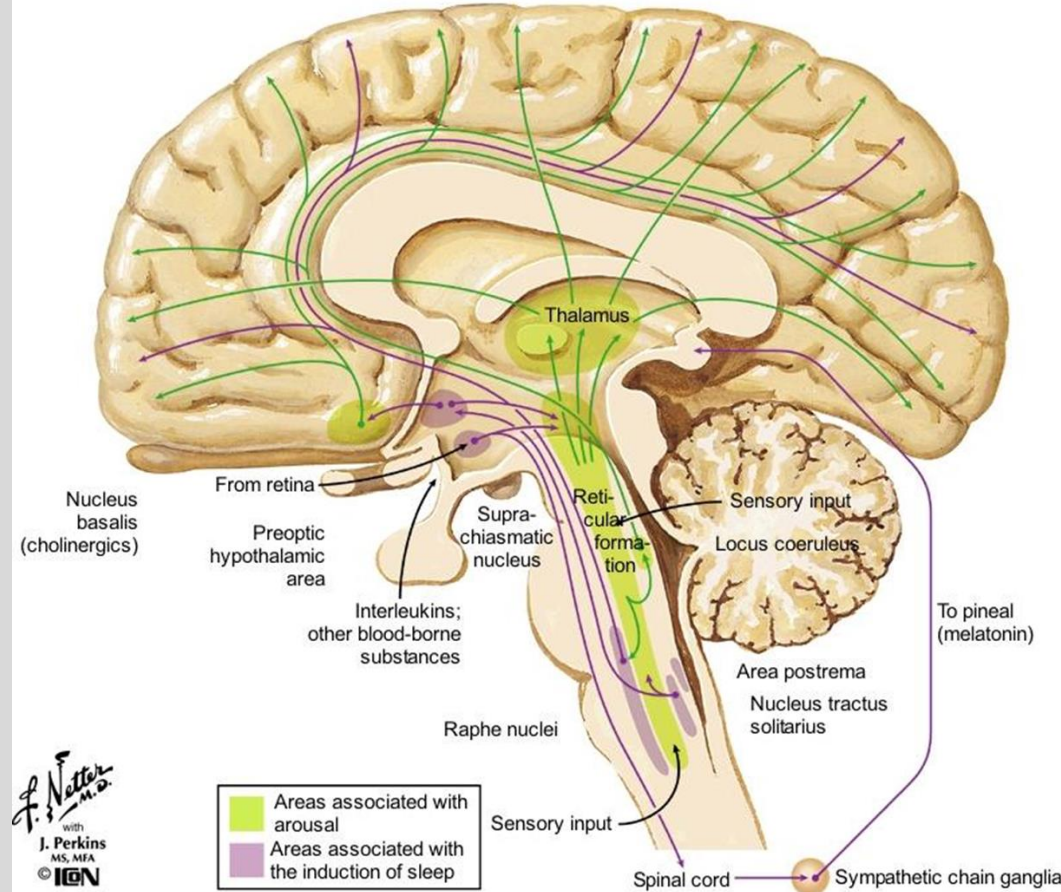
Fig. 3. T2-weighted axial MRI showing massive and bilateral pontine infarction.

The Diving Bell and the Butterfly



* Contrast with ALS, death of upper motor neurons

Sleep-Wakefulness Control

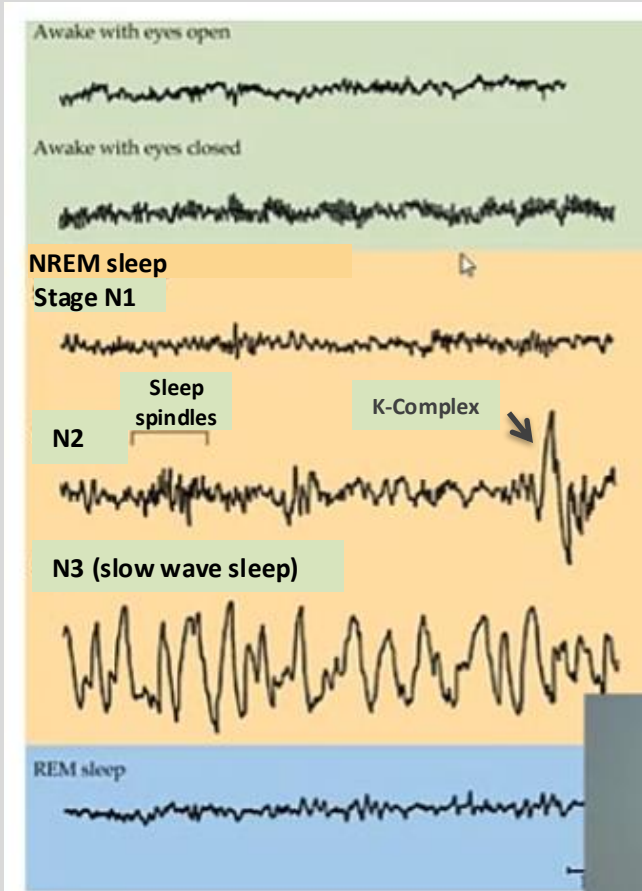


J. Natter
with
J. Perkins
MS, MFA
© ION

Definition of sleep:

- Reduced motor activity
- Decreased response to stimulation
- Stereotypic postures
- Relatively easy reversibility

EEG is associated with different stages of sleep and wakefulness



- **Awake**

- **Awake with eyes open:** high frequency, low amplitude **beta (β)** waves
- **Relaxed wakefulness (eyes shut)** shows rhythmic 8–12-Hz **alpha (α)** waves

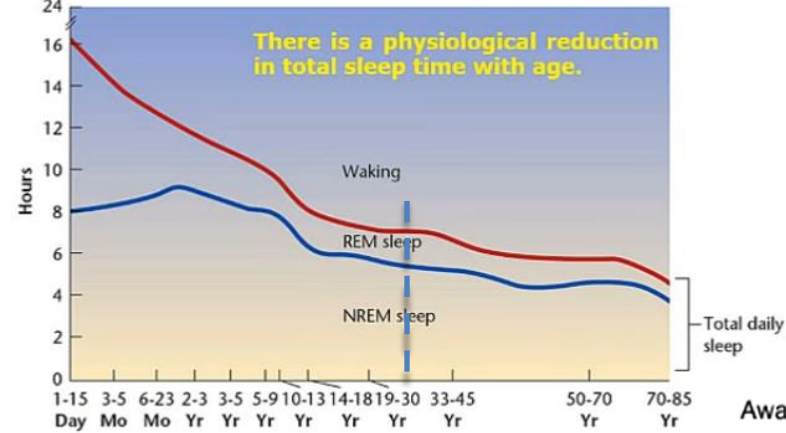
- **NREM (Non-REM) sleep**

- **Stage N1** shows mixed frequencies, especially 3–7-Hz **theta (θ)** waves
- **Stage N2** shows 12–14-Hz **sleep spindles** and **K-complexes**
- **Stage N3 (slow wave sleep)** shows large-amplitude ($>75 \mu\text{V}$) 0.5–2-Hz **delta (δ)** waves.

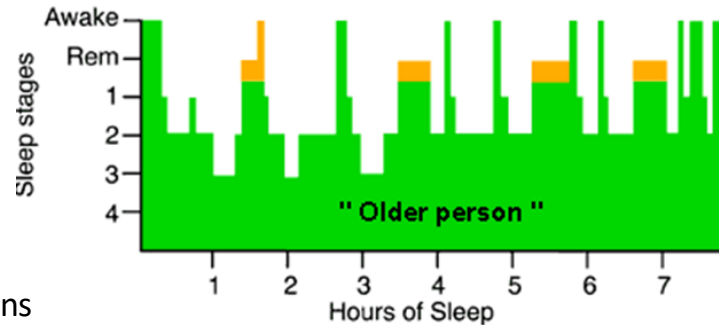
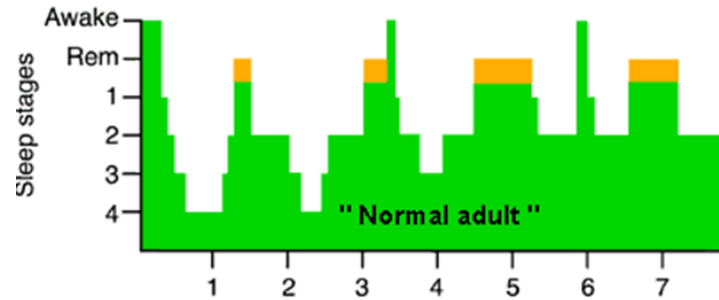
- **REM (Rapid Eye Movement) sleep**

- shows low-amplitude, mixed frequencies with saw-tooth waves, **close to beta (β)** waves

Changes in sleep cycle with age



Hypnogram



Old stage definitions

Physiological Correlates During Sleep

NEW YORK INSTITUTE
OF TECHNOLOGY

College of Osteopathic
Medicine

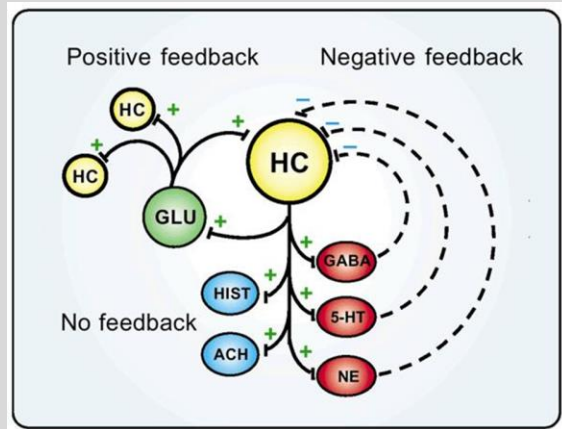
	<u>SWS Sleep</u>	<u>REM Sleep</u>
Heart Rate	↓ from awake	↑ from SWS
Blood Press.	↓ from awake	↑ from SWS
Respiratory Rate	↓ from awake	↑ from SWS
Blood Flow to Brain	↓ from awake	↑ from SWS**
Para (Gut Motility)	↑ from awake	similar to SWS
Body Temperature	↓ from awake	↓ from SWS
Muscle Tone	↓ from awake	↓ from SWS
Dreams	Very Few	Most Dreaming
Eye Movements	Slow Moving	Rapid Eye Move

Neurochemistry of Sleep

NEW YORK INSTITUTE
OF TECHNOLOGY

College of Osteopathic
Medicine

- Neurotransmitters promoting wakefulness:
 - Glutamate: universal
 - Serotonin neurons: *Raphe nuclei*
 - Norepinephrine neurons: *Locus coeruleus*
 - Acetylcholine neurons: *basal forebrain and dorsal tegmentum*
 - Histamine neurons: *Tuberomammillary nucleus*
 - **Hypocretin (Orexin)** neurons: **Lateral hypothalamus**
 - Narcolepsy: low hypocretin in the brain

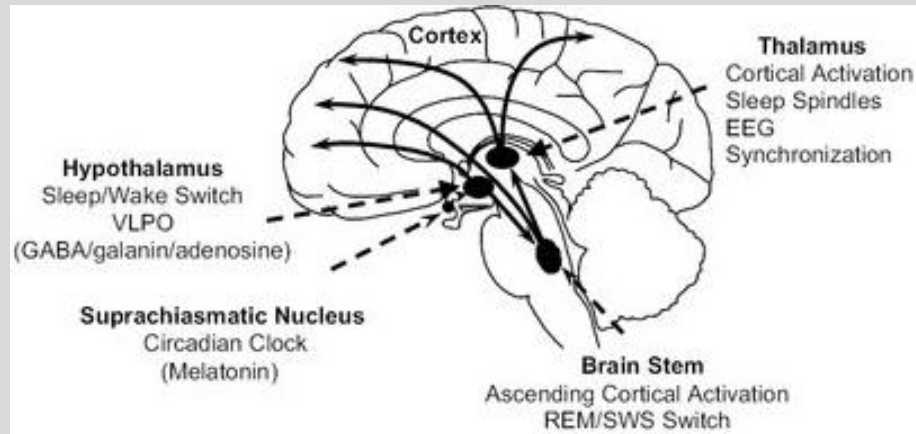


Neurochemistry of Sleep

- Neurotransmitters promoting sleep:
 - Gamma-aminobutyric acid (**GABA**): *basal forebrain (medial septum and diagonal band) and hypothalamus (preoptic area)*; e.g., benzos, barbiturates
 - **Melatonin**: *suprachiasmatic nucleus → pineal gland*

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine



VLPO: ventrolateral preoptic nucleus

Firing Patterns of Modulatory Neurons During Non-REM and REM Sleep

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

Nucleus	Transmitter	Wakefulness	Non-REM Sleep	REM Sleep
Preoptic area	GABA	-	++	++
Lateral Hypothalamus	Orexin/Hypocretin	++	-	-
Tuberomammillary nucleus	Histamine	++	-	-
Raphe nuclei	Serotonin	++	+	-
Locus ceruleus	Norepinephrine	++	+	-
Basal forebrain	Acetylcholine	++	+	++

Narcolepsy Treatment



Stimulants: to treat excessive daytime sleepiness

- Dextroamphetamine – sympathemimetic: promoting release and inhibiting reuptake of DA, NE, 5-HT
- Methylphenidate – NDRI (norepinephrine/dopamine reuptake inhibitor) → inhibits reuptake of NE and DA

Drug used as sleep aids

	Zaleplon (Sonata®)	Potentiate GABAs ability to open the GABA _A chloride channel (increase channel opening frequency) very short acting	Sedation, Dizziness
	Zolpidem (Ambien®)	Potentiate GABAs ability to open the GABA _A chloride channel (increase channel opening frequency) short acting	Sedation, Dizziness
	Eszopiclone (Lunesta®)	Potentiate GABAs ability to open the GABA _A chloride channel (increase channel opening frequency) long acting	Sedation, Dizziness, Somnolence
	Chloral hydrate	Increase duration of GABA _A chloride channel openings, at high concentrations can open directly	Sudden acute intoxication-death, Dependence, Severe withdrawal, Liver damage
	Buspirone (BuSpar®)	5-HT _{1A} receptor partial agonist	Sedation, Ataxia, Should not be taken with MAOIs, Withdrawal
	Suvorexant (Belsomra®)	Orexin receptor antagonist	Sedation, mental cloudiness, CYP3A4
	Ramelteon (Rozerem®)	Melatonin MT ₁ and MT ₂ receptor agonist	hyperprolactinaemia, Possible teratogen, uses CYP1A2, CYP3A4, CYP 2C9

Functions of Reticular Formation

**NEW YORK INSTITUTE
OF TECHNOLOGY**

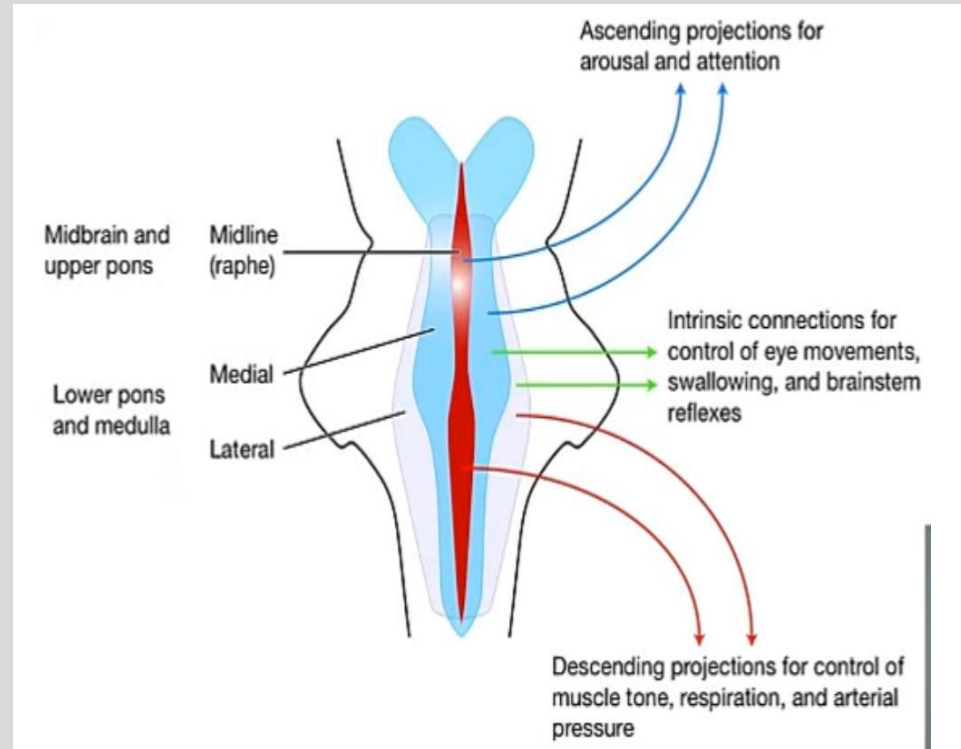
College of Osteopathic
Medicine

Ascending:

- Arousal
- Attention
- Sleep

Descending:

- Motor Functions
- Micturition
- Pain
- Cardiac Reflexes
- Respiration



Different levels of damage causes different rigidity

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

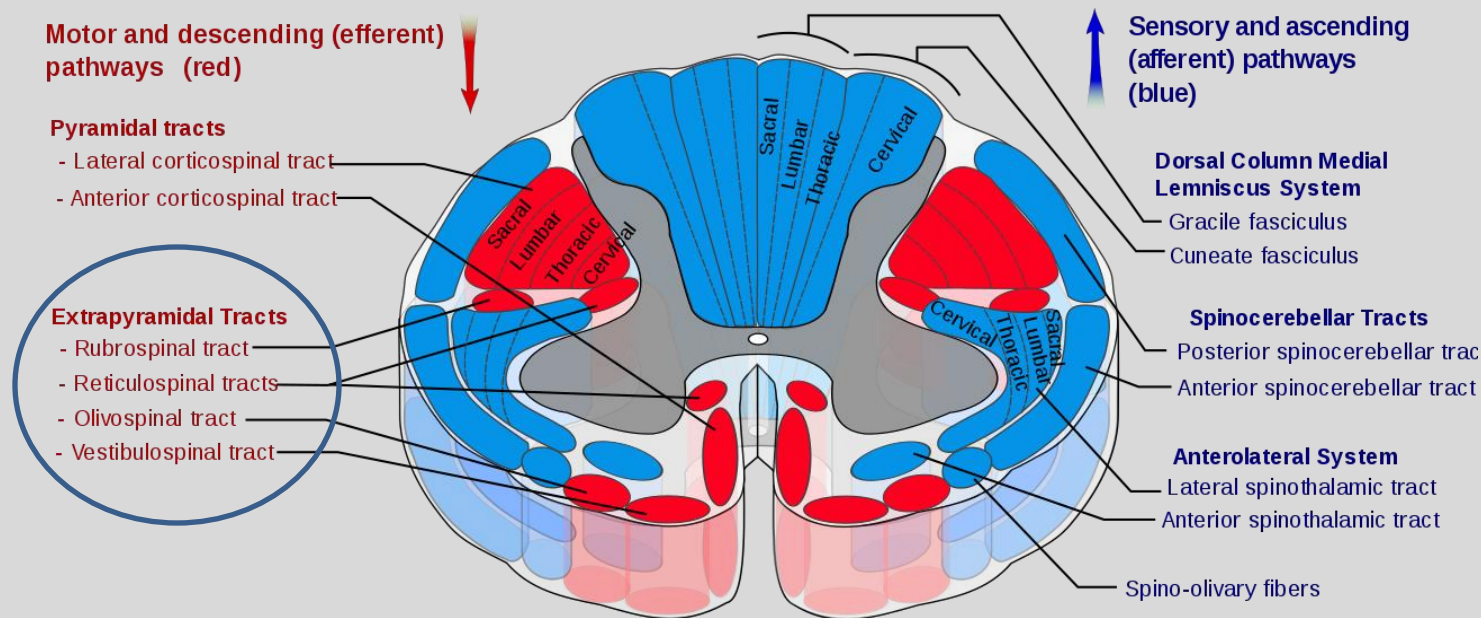


<https://www.youtube.com/watch?v=yZUE2Dvf1Q4>

Descending Motor Control

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine



Decorticate vs. Decerebrate Rigidity

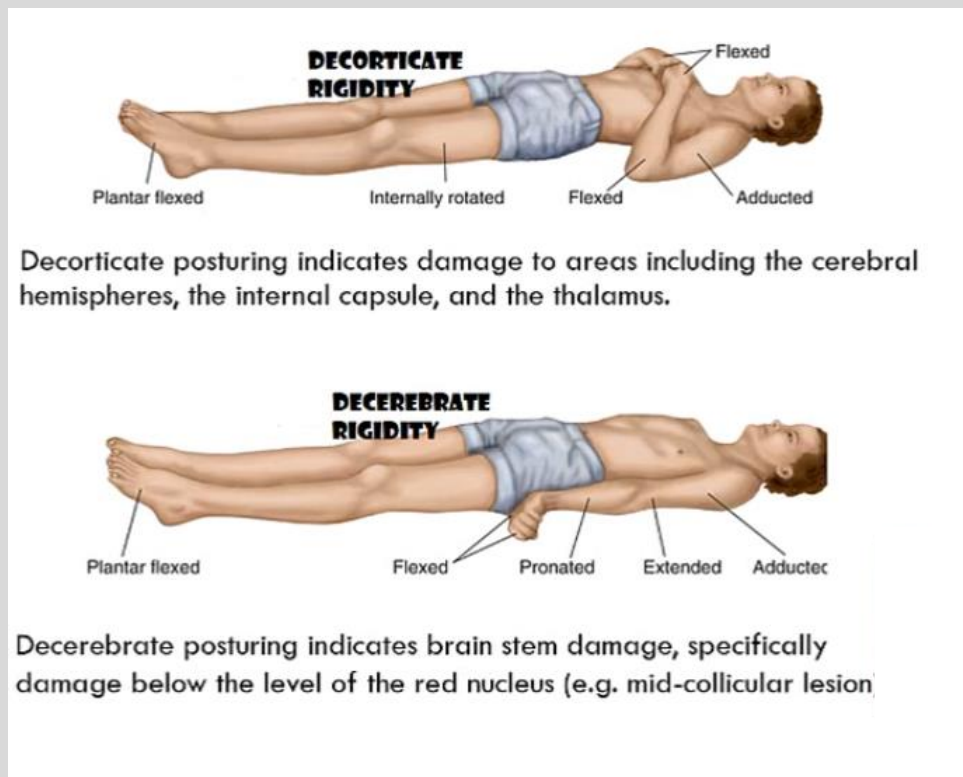
**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

Decorticate rigidity: When damage **above red nucleus**, brain stem motor center **rubrospinal** tract is intact- **flexor** response to **upper** extremities, **extension** for **lower** extremities

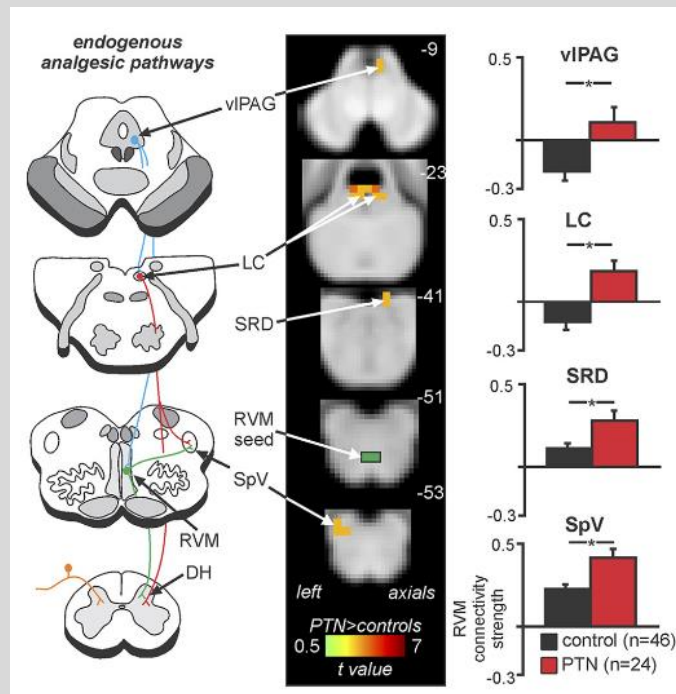
Decerebrate rigidity: When damage **at or below red nucleus**, brain stem motor center is damaged - **extensor** response for all **4 limbs**

-**Vestibulospinal** and **reticulospinal** tracts dominating



Descending Pain Modulation

- Stimulation of **periaqueductal gray** matter (**PAG**) excites **locus coeruleus** (NE), and rostral ventromedial medulla (**RVM**) to inhibit pain
- These project to the dorsal horn (**substantia gelatinosa**) and activate **enkephalin**-containing interneurons
- PAG receives input from spinomesencephalic tract, i.e. ascending pain information drives descending pain inhibition
- PAG has abundant opioid receptors.
- Chronic neuropathic pain patients engage this pathway → descending pain facilitation



vPAG: ventrolateral PAG

LC: locus coeruleus

SRD: subnucleus reticularis dorsalis

RVM: rostral ventromedial medulla

SpV: spinal trigeminal nucleus

DH: dorsal horn

PTN: painful trigeminal neuropathy

Mills et al., J. Neurosci., 2018, 38(2):465

<https://www.jneurosci.org/content/jneuro/38/2/465.full.pdf>

Chronic Pain Therapeutics

Chronic pain patients - diminished descending pain inhibition

Current first line treatments for chronic pain engage descending pain inhibitory system:

- Amine antidepressants:
 - TCAs (tricyclic antidepressants):
 - Amitriptyline - serotonin norepinephrine reuptake inhibitor (SSNRI);
 - Desipramine: norepinephrine-selective reuptake inhibitor (NSRI), weaker therapeutic effect
 - Floxetine: selective serotonin reuptake inhibitor (SSRI), high dose can treat fibromyalgia
 - Duloxetine: SNRI

The efficiency of conditioned pain modulation (CPM, “pain inhibits pain”) predicts the efficacy of duloxetine.

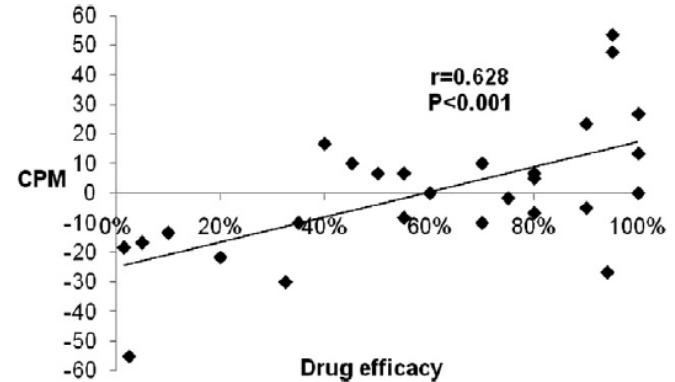
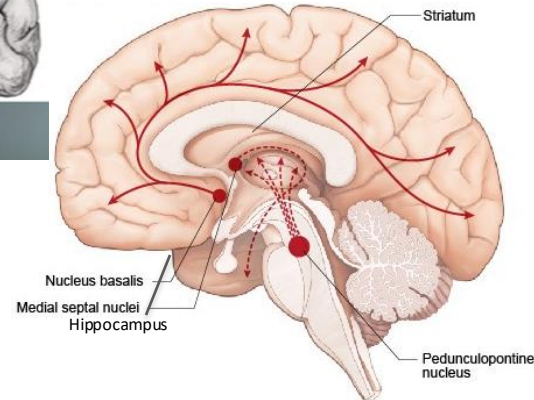
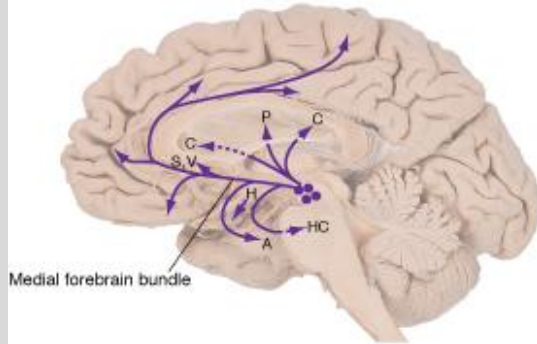
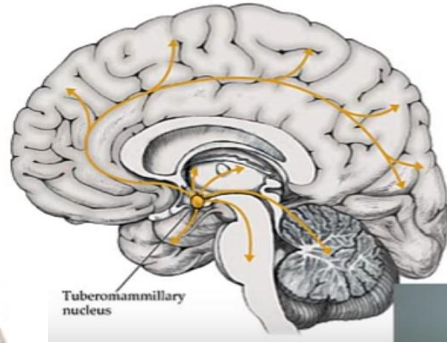
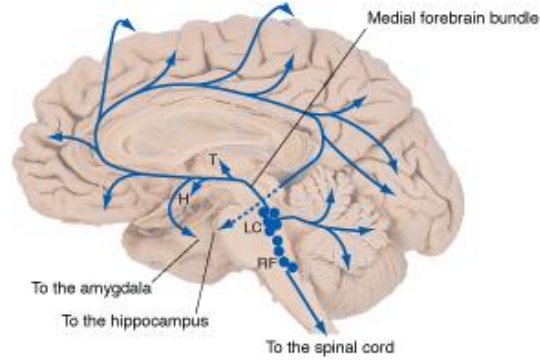
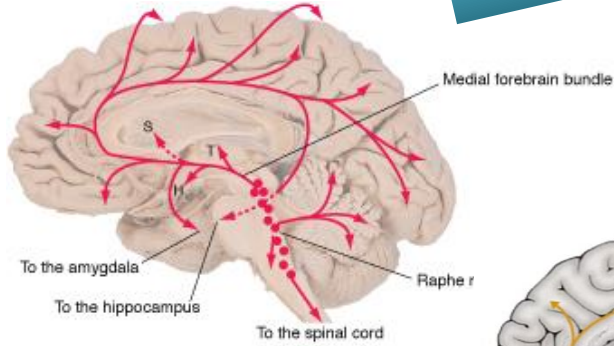


Fig. 1. Drug efficacy and pretreatment CPM. Patients with less efficient CPM (positive scores) reported higher drug efficacy and vice versa.

Yamitsky, D., et al. (2012), *Pain* **153**(6): 1193-1198

Summary

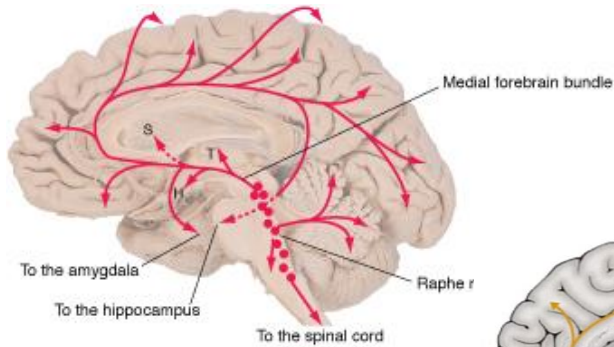
Which pathway?
Neurotransmitter?



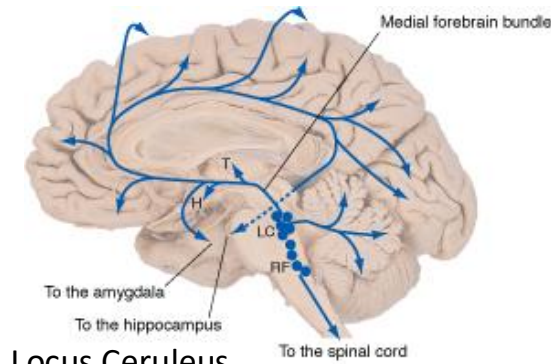
YORK INSTITUTE
CHNOLOGY

e of Osteopathic
ne

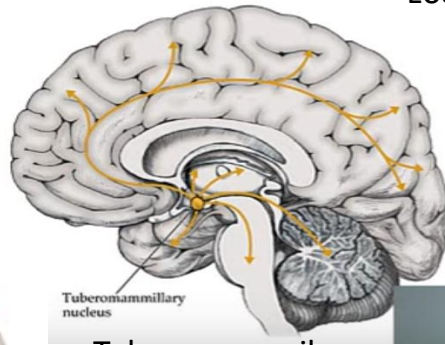
Summary



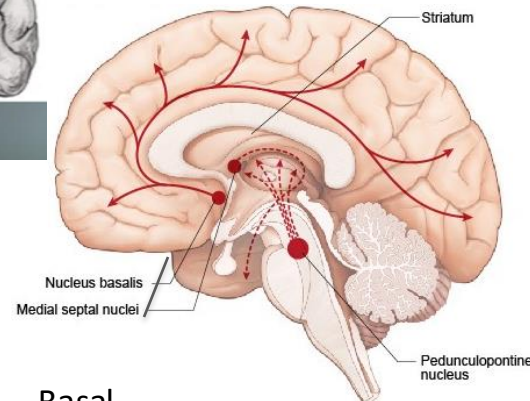
Raphe Nuclei (5-HT)



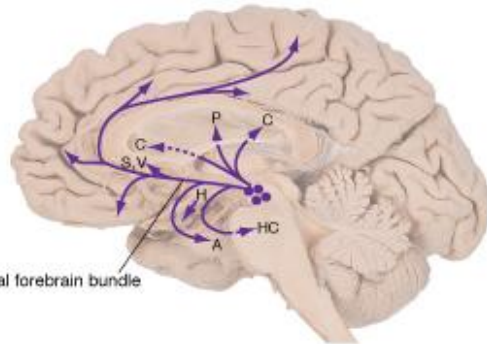
Locus Ceruleus (NE)



Tuberomammillary nucleus (His)



Basal forebrain/Pedunculo pontine (ACh)



Ventral tegmental & Substantia Nigra (DA)

Fill in the blank

Summary

NEW YORK INSTITUTE
OF TECHNOLOGY

College of Osteopathic
Medicine

Functions	<u>Anatomical area</u>	<u>Neurotransmitter(s)</u> (incomplete list)
<u>Arousal</u> , attention		
Sleep <u>—</u>		
Motor control <u>—</u>		
<u>Pain</u> modulation		
<u>Bladder</u> control		
<u>Respiration</u>		

Summary

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

Functions	Anatomical area	Neurotransmitter(s) (incomplete list)
Arousal, attention	Ascending reticular activating system (ARAS)	ACh, NE, DA, 5-HT, His
Sleep	Reticular formation, hypothalamus	ACh, His, NE, 5-HT, GABA, hypocretin
Motor control	Pedunculopontine tegmental nuclei (PPT), red nucleus, extrapyramidal tracts	ACh, DA, Glu, GABA
Pain modulation	Periaqueductal gray (PAG), Raphe magnus, locus coeruleus	Enkephalin, 5-HT, NE
Bladder control	Pontine micturition center, PAG	Glu, GABA
Respiration	Medial parabrachial nucleus (breathing pattern)	Glu

Fill in the blank:

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

_____ is produced by neurons found in the locus coeruleus.

_____ is produced by neurons of the raphe nuclei.

Low levels of _____ is associated with severe depression and insomnia.

_____ is produced by neurons found in the basal nucleus of Meynert.

_____ is the neurotransmitter of the Renshaw cells.

_____ is found in high concentration in the pars compacta of the substantia nigra and ventral tegmental area of the mesencephalon.

_____ is the neurotransmitter of the mesolimbic pathway.

Decorticate rigidity is due to damage of the motor neurons _____ (above, at, below) the level of _____, with the exaggerated _____ response at the upper extremities and _____ response at the lower extremities.

Fill in the blank:

NE is produced by neurons found in the locus ceruleus.

5-HT is produced by neurons of the raphe nuclei.

Low levels of 5-HT is associated with severe depression and insomnia.

ACh is produced by neurons found in the basal nucleus of Meynert.

Gly is the neurotransmitter of the Renshaw cells.

DA is found in high concentration in the pars compacta of the substantia nigra and ventral tegmental area of the mesencephalon.

DA is the neurotransmitter of the mesolimbic pathway.

Decorticate rigidity is due to damage of the motor neurons above (above, at, below) the level of red nucleus, with the exaggerated flexor response at the upper extremities and extensor response at the lower extremities.

College of Osteopathic
Medicine

A photograph of a young child with dark hair, wearing a striped long-sleeved shirt and a red bow tie. The child is looking upwards and pointing their right index finger towards the top right. The background is white and filled with several large, black question marks of varying sizes, suggesting a state of confusion or inquiry.

Email: jennifer.xie@nyit.edu